

FreeSpiral: AI-Guided GFP Design Report

2026 Synthetic Biology Innovation Competition
Protein Design Challenge

Team: FreeSpiral | Date: June 23, 2026

Score = Relative Brightness × Thermal Stability Retention

Executive Summary

We present **FreeSpiral**, an AI-guided protein design pipeline integrating multi-source data analysis, literature-backed rational engineering, and LLM reasoning to generate six optimized GFP variants for the 2026 Protein Design Challenge.

1. Design Methodology

Data-Driven Analysis: Parsed 141,572 mutation-brightness measurements from GFP_data.xlsx. Top mutation K157G gives +2.48× brightness. Analyzed 20 previous top-10 sequences.

Literature Integration: Superfolder GFP (Pédelacq 2006), TGP (Close 2015), Nature fitness landscape (Sarkisyan 2016), StayGold (Ivorra-Molla 2024).

2. Final 6 Sequences

Seq	Strategy	Key Mutations	RMSD@72C
Seq1	Champion	S65T,S72A,Q80R,Y143G,S147P,Q157G,V163A,I167T,I171V,S202D	1.804nm
Seq2	BrightStar	S30R,S65T,S72A,Q80R,Y143G,S147P,Q157G,V163A,I167T,I171V	1.824nm
Seq3	sfGFP+	S72A,Y143G,S147P,Q157G,I167T,I171V on sfGFP	1.806nm
Seq4	sfGFP-Stable	L18E,S147P,Q157G,I167T,I171V on sfGFP	1.801nm
Seq5	Conservative	S30R,L64F,S65T,S72A,Q80R,S147P,V163A,I167T,I171V,S202D	1.837nm
Seq6	Evolved	S65T,S72A,Q80R,Y143G,Q157G,I167T,I171V,T203I	1.812nm

3. LLM Agent Decision Tree

- INPUT: Competition materials (datasets, sequences, exclusion list, papers)
- STEP 1: Parse 141K brightness entries, 20 winner sequences, 135K exclusion entries
- STEP 2: Rank mutations by fold-change, cross-reference winners, verify mapping
- STEP 3: Extract validated mutations from literature
- STEP 4: Create 6 strategies balancing brightness vs stability
- STEP 5: Generate on server with PDBFixer/OpenMM MD validation
- STEP 6: Validate length, start codon, amino acids, exclusion list
- OUTPUT: 6 validated sequences in submission.csv

4. Validation & MD Results

- ✓ Length: All 238-239 aa (within 220-250)
- ✓ Start: All begin with Methionine (M)
- ✓ No stop codons, standard amino acids only
- ✓ Exclusion list: No matches in 135,414 entries
- ✓ MD at 72°C (0.5ns): Seq4 best stability (1.801nm), all maintain fold
- ✓ Simulation: 2× RTX 3090, OpenMM 8.5.2, Amber14, 34-43K atoms/system

5. Repository

GitHub: github.com/MengDi-web/FreeSpiral-2026-Protein-Design

References

- [1] Pédelacq et al. (2006) Nat. Biotechnol.
- [2] Close et al. (2015) Proteins
- [3] Sarkisyan et al. (2016) Nature
- [4] Ivorra-Molla et al. (2024) Nat. Biotechnol.